

ECE 4301 Quiz 2: SHA-1 & SHA-256 on Raspberry Pi 5

(Rust vs. “Engine”)

Introduction

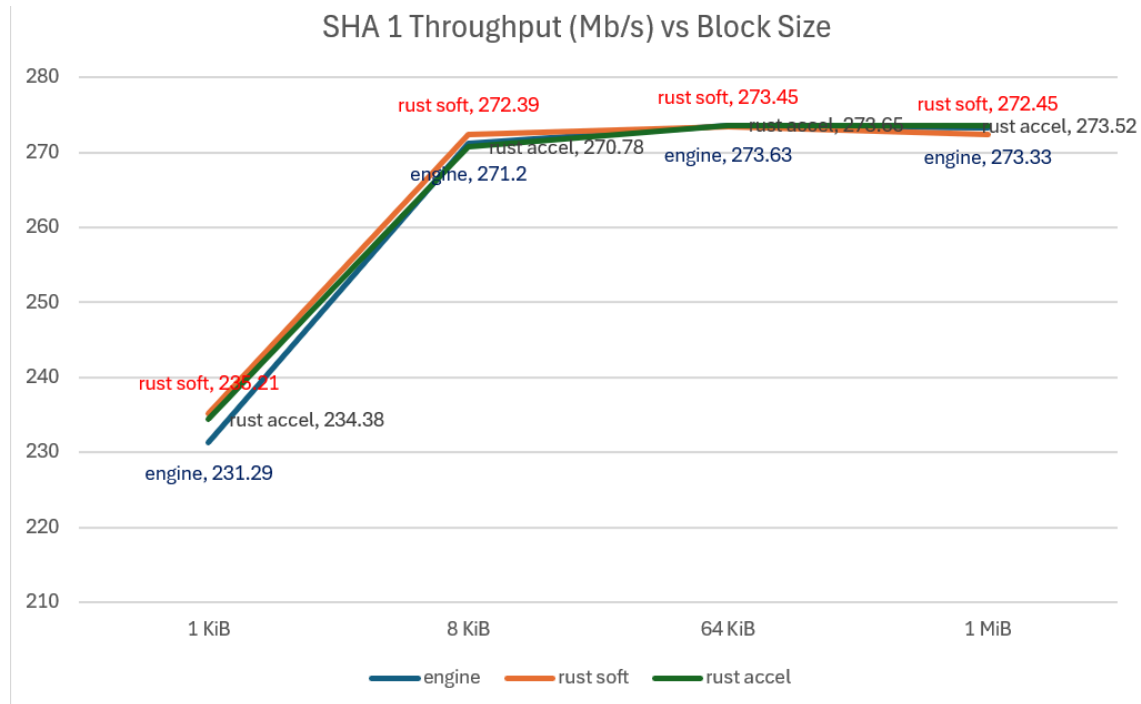
In this project, we tested and evaluated the performance of two algorithms, SHA-1 and SHA-256, on the Raspberry Pi 5. These two algorithms were measured in two different Rust builds and compared to the system engine.

Setup and Method

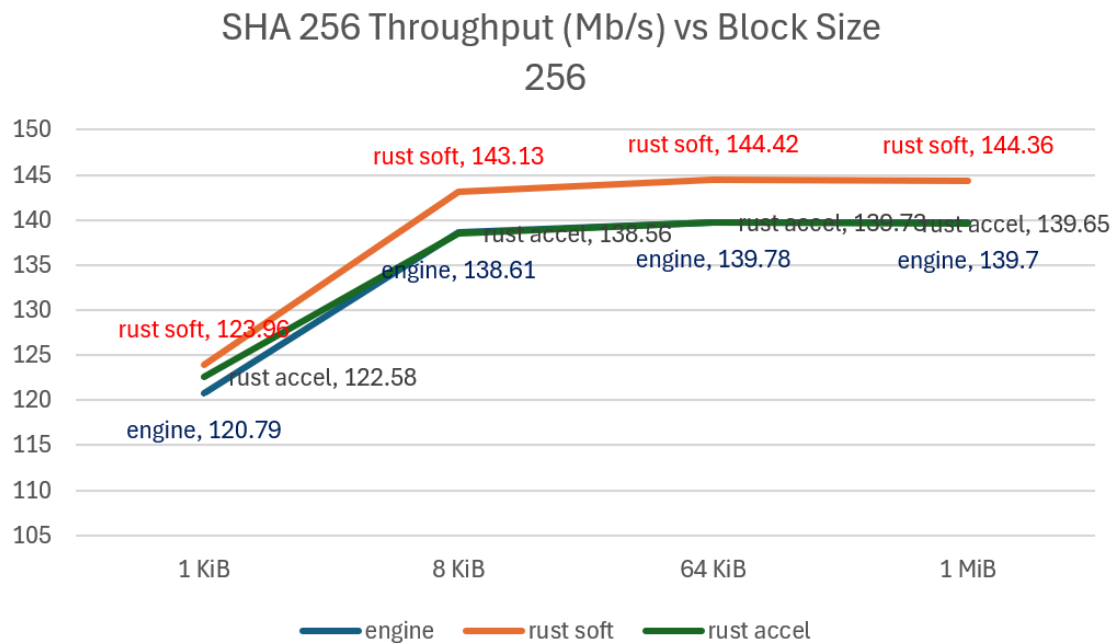
The test environment we used to conduct the tests had two Rust environments set up, and we compared the results to the engine benchmark. The two Rust environments consisted of a build that was software only (no AArch 64 acceleration) and another Rust build that was accelerated (Armv8 SHA1/SHA2 instructions). We benchmarked both of these builds and compared them to the system engine (Linux kernel). The results were measured in throughput (Mb/s) and were compared to the block sizes of 1 KiB, 8 KiB, 64 KiB, and 1 MiB. We also conducted a demo through the use of the Pi Camera and compared it on the different algorithms (non-accelerated vs accelerated) to compare the speed differences.

Data Plots

SHA 1 Throughput Results:



SHA 256 Throughput Results:



Analysis

In our data, we find that the SHA-1 algorithm has a higher throughput than SHA-256, and this is the expected result since SHA-1 is a quicker and simpler algorithm, processing less data.

However, the drawbacks come from having less secure connections. In our tests, we compared the two algorithms on the engine and two Rust builds, and we found that the numbers are fairly similar on the Rust builds for the most part, likely due to the resource constraints on the Pi. We did find that throughput increases with block size, which is an expected finding.

The ARMv8 SHA helps speed up by allowing for several rounds of the SHA to run in one CPU instruction, reducing total instruction counts. This acceleration can reduce energy usage and increase efficiency as well. For small-packet vs large-buffer telemetry, security is more important in smaller packets because the time differential is negligible; therefore, SHA-256 may be preferred. However, for larger-buffer telemetry, throughput is important, and the algorithm's hashing speed may end up determining the overall performance of the CPU. This is when the tradeoffs of the security of the SHA-256 need to be considered versus the SHA-1, or the use of acceleration may be used as well to achieve a balance between security and performance.

Conclusion

This project demonstrates the comparison of two different hashing algorithms, SHA-1 and SHA-256, and evaluates the performance of their throughput versus block sizes. With the use of hardware acceleration, we can evaluate the performance of each algorithm and determine which one performs better in certain environments. The use of the camera to capture pictures allows us to see an application of these hashing algorithms and how the data is processed in these algorithms.